Optimizing the Egyptian Premier League Schedule

A Mathematical Optimization Approach using Integer Linear Programming

01 Introduction

Introduction To The Project And Problem Overview

The origin of the story

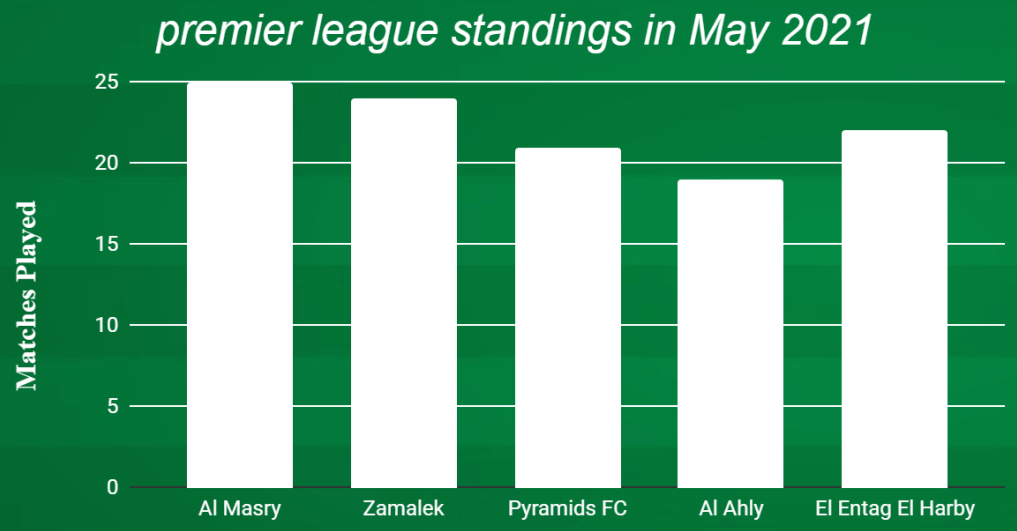
- For years, league scheduling relied on manual planning, quietly accumulating issues that became impossible to ignore during Egypt’s emergency hosting of AFCON 2019.This forced a significant disruption to the standard calendar.
- The league is currently trapped in a "Domino Effect" of delayed starts and compressed schedules, which results in:
  - Fixture Congestion: Teams facing impossible turnaround time
  - Uneven Rest: Disparities in recovery time between competing clubs.
  - Frequent Postponements: A chaotic schedule that changes frequently.

The Aggravator (COVID-19)

- **Global Disruption:** COVID-19 disrupted global sport, resulting in further delays.

- **This accumulation pushed season start dates as far back as December, creating a time deficit that was never recovered.**

Visualizing the Disparity



This chart illustrates the 2020/2021 problem's peak, comparing the disparity in matches played by five teams at the same point in the season.

2024/2025 Exceptional Format

- **The Qualifier**: All 18 teams compete in a single round-robin format (17 matches per team), with no return legs.
- **The Split:**
  - **Top 9:** Compete for the league title and continental qualification.
  - **Bottom 9:** Compete to avoid relegation.

Our Goal

- **Structured Framework:** Model the league using Integer Linear Programming (ILP).
- **Time Management:** Fit the standard 34-round season strictly between August and May.
- **Constraint Satisfaction:** Respect CAF commitments, Security refusals, and FIFA breaks.
- **Fairness:** Balance Home/Away streaks and travel distances.

The Proposed Solution

An Optimization-Based System.

- **Input** — Structured Data (Teams, Stadiums, Rules)
- **Model** — Mathematical Constraints & Objective Functions
- **Output** — An efficient schedule that balances efficiency and fairness.

Model

Objectives

Maximize Commercial Value

- Important matches in prime-time weekend

Maximize Fairness

- Enforce **simultaneous kick-offs** for the final critical rounds
- Balance **rest days** between opponents to ensure equity

Minimize Travel Fatigue (TTP)

Constraints

Stadium Sharing

- No simultaneous matches allowed for teams sharing a home venue

Continental Blockers

- Mandatory ± 3 **day buffer** around any CAF

Security Matrix

- High-risk matchups must be in neutral venues

League Structure

- Every team plays exactly ***one match per Game Week*** (Double Round Robin integrity)

02 Background

Background And Existing Work

Sports Scheduling as an Optimization Problem

Sports scheduling is a subfield of Operations Research and **Combinatorial Optimization**

NP-hard problem

Complexity grows exponentially with:

- number of teams
- number of constraints

DRR format

Standard football leagues follow a **Double Round Robin**

- Each team plays every other team **twice** (home & away)
- For *n* teams → **n(n−1)** total matches

In the EPL:

Egyptian Premier League has:

- **18 teams**
- **306 matches**
- **34 rounds**

Fairness & Travel

Patterns & Breaks

- **Pattern** is a sequence of Home (H) and Away (A) matches
- **Break** = H–H or A–A
- Away breaks = unfair & exhausting
- Perfect alternation (H–A–H–A) for all teams is **mathematically impossible**

Travel & TTP

Traveling Tournament Problem (TTP) :

- Introduced to **minimize total travel distance**
- Especially important in geographically spread leagues
- Travel fatigue directly affects performance and recovery

Chile Case Study

In 2005, the Chilean First Division switched from manual/random draws to ILP Optimization and here's the results:

36% Increase in attendance.

2x Increase in ticket revenue.

45% Reduced travel costs and fairer distribution.

The Egyptian Gap

Why can't we just copy?

Security Approvals

Specific match-ups are banned in specific cities

Shared Resources

Multiple teams share "Cairo International" and "Borg Al-Arab," creating bottlenecks

CAF Conflict

Al Ahly & Zamalek schedules are fluid, breaking standard mathematical patterns.

03 PROJECT DATA

Data And Preprocessing

Data Collection

- **Sources:** Automated Web Scraping (Python) from "*Yallakora*" & "*Transfermarkt*", with additional data manually collected from articles. **Why?** To ensure real-time accuracy and consistency with official records.
- **Integration:** Exported data into spreadsheets for data processing
- **Historical Benchmarking:** Collected past season data to measure improvements and efficiency.

Data Architecture

Teams

Administrative details & Home venues

Stadiums

Locations, Capacities & Floodlight availability

Calendar

Every available "Slot" (Date + Time)

Security Matrix

A rigid list of Forbidden Matchups & Forced Venues

Continental Blockers

- CAF Schedules & Travel Buffers
- FIFA Days

Distance Matrix

KM distance between every 2 nodes on a network (for travel costs)

Data Preprocessing

Standardization

- Converted inconsistent date formats from different websites
- All dates are now in a standard **MM/DD/YYYY** format for system compatibility

Unification

- Fixed inconsistent team names
- Every team has a unique **Team ID** (e.g., "AHL") that works across all files

Linking

- Connected the separate Excel sheets using the new IDs
- Transformed "flat" files into a **Connected Schema.**

Data Modeling

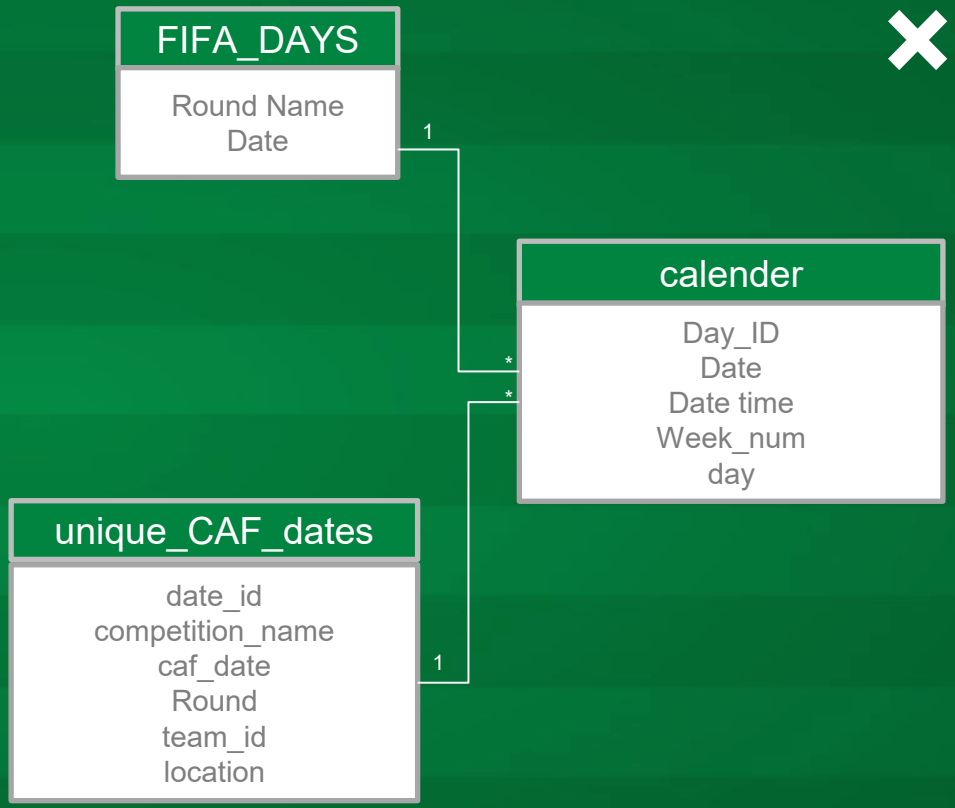
Infrastructure Model

- **Purpose:** Connects every team to their specific home venue.
- **How it Works:** Uses a "Many-to-One" link because multiple teams can share one stadium.
- **Travel Calculation:** Links stadiums to a Distance Matrix to automatically calculate travel costs.

Data Modeling

Temporal Model

- **Purpose:** A "Star Schema" centered around the main Calendar.
- **How it Works:** Connects the master calendar to FIFA and CAF lists.
- **Blocking:** Automatically "blocks" league matches during international or continental dates.



Conclusion Of Phase I

- Clearly defined scheduling inefficiencies in the Egyptian League
- Identified operational, security, and continental constraints
- Built a structured, connected data framework and validated datasets for optimization

- Next Phase: Formulating and implementing the model
- Translating constraints into equations
- Generating an efficient, fair, and feasible schedule

References

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Thank You